TEXORA · COURSE NOTES · LECTURE SHEET

Rotational Inertia of Solid Objects

Moment of inertia from first principles — the two axis theorems, standard results for rods, rings, discs and spheres, rotational energy, and rolling motion.

PHY-1103 · PHYSICS-I · TOPIC 1 · LEVEL 1 TERM 1

1. Why rotation needs its own inertia

Mass measures how a body resists a change in its linear motion. A spinning body resists in a different way: push on a flywheel near its axle and it barely responds, push near its rim and it answers readily — the same force is more effective the farther it acts from the axis. The quantity that captures this is the **moment of inertia**, I. For a system of particles rotating about a fixed axis, each particle of mass m_i at distance r_i from the axis contributes $m_i\, r_i^2$, and

$$I = \Sigma\, m_i\, r_i^2 \quad \rightarrow \quad \text{for a continuous body: } I = \int r^2\, dm$$

Three facts follow directly from the definition. I depends on the **total mass**; it depends far more strongly on **how the mass is distributed** (distance enters squared — moving mass from the rim to the hub slashes I); and it depends on **which axis** is chosen, so a moment of inertia is meaningless until the axis is named. Its unit is kg m². A useful shorthand is the **radius of gyration** K, defined by $I = MK^2$: the whole mass, imagined concentrated at distance K from the axis, would have the same I.

2. The two axis theorems

Parallel axis theorem (Steiner). The moment of inertia about any axis equals the moment about a parallel axis through the centre of mass, plus Md^2, where d is the distance between the axes:

$$I = I_{cm} + M\, d^2$$

Perpendicular axis theorem (flat bodies only). For a plane lamina, the moment of inertia about an axis perpendicular to its plane equals the sum of the moments about two perpendicular axes in its plane through the same point:

$$I_z = I_x + I_y$$

These two theorems generate nearly every standard result from a few known ones — as the worked example below shows.

3. Standard results

BODY	AXIS	MOMENT OF INERTIA
Thin rod, length L	Through centre, perpendicular to rod	$(1/12)\ ML^2$
Thin rod, length L	Through one end, perpendicular to rod	$(1/3)\ ML^2$
Ring / thin hoop, radius R	Through centre, perpendicular to plane	MR^2
Disc / solid cylinder, radius R	Through centre, perpendicular to plane (= cylinder axis)	$\frac{1}{2}\ MR^2$
Solid sphere, radius R	Through centre (any diameter)	$(2/5)\ MR^2$
Hollow sphere (thin shell), radius R	Through centre	$(2/3)\ MR^2$
Rectangular plate, sides a, b	Through centre, perpendicular to plate	$(M/12)(a^2 + b^2)$

The ordering teaches the physics: for the same mass and radius, ring (all mass at R) > hollow sphere > disc > solid sphere — I falls as mass moves toward the axis. A quick derivation worth knowing: a **disc** is a stack of rings; a ring of radius r and width dr has mass $dm = (M/\pi R^2)\cdot 2\pi r\, dr = (2M/R^2)\, r\, dr$, so $I = \int_0^R r^2\, dm = \int_0^R (2M/R^2)\, r^3\, dr = (2M/R^2)(R^4/4) = \frac{1}{2}\, MR^2$. The **rod about its centre** follows the same pattern: with linear density $\lambda = M/L$, $I = \int_{-L/2}^{L/2} \lambda x^2\, dx = \lambda\, L^3/12 = ML^2/12$.

WORKED EXAMPLE — THE ROD ABOUT ITS END, BY STEINER

Given the rod about its centre, $I_{cm} = ML^2/12$. The end axis is parallel, at distance $d = L/2$. Then $I = ML^2/12 + M(L/2)^2 = ML^2/12 + ML^2/4 = ML^2/12 + 3ML^2/12 =$ **$(1/3)\ ML^2$** — exactly the table value, obtained without a new integral. This is the daily use of the parallel axis theorem.

4. Rotation dynamics: torque, energy, angular momentum

With I defined, the whole of linear dynamics translates term by term:

QUANTITY	LINEAR	ROTATIONAL
Inertia	mass m	moment of inertia I

Force law	F = ma	τ = Iα (torque = I × angular acceleration)
Momentum	p = mv	L = Iω (angular momentum)
Kinetic energy	½ mv²	½ Iω²
Work / power	W = F·s, P = Fv	W = τ·θ, P = τω

WORKED EXAMPLE — ENERGY STORED IN A FLYWHEEL

A solid disc flywheel of mass 10 kg and radius 0.50 m spins at 20 rad s^{-1}. I = ½ MR^2 = ½ × 10 × 0.25 = 1.25 kg m^2. Stored energy = ½ $I\omega^2$ = ½ × 1.25 × 400 = **250 J**. This is why flywheels smooth machines: they bank rotational energy at speed and pay it out when the load surges — the principle of the loom's drive and of every engine's flywheel.

5. Rolling motion

A body rolling without slipping combines translation of its centre with rotation about it, linked by $v = \omega R$. Its kinetic energy is the sum of both parts, which collapses neatly using $I = MK^2$:

$$KE = \tfrac{1}{2} Mv^2 + \tfrac{1}{2} I\omega^2 = \tfrac{1}{2} Mv^2 \left(1 + K^2/R^2 \right)$$

Released down an incline of angle θ, a rolling body accelerates at

$$a = g \sin\theta / \left(1 + K^2/R^2 \right)$$

— slower than a sliding block (g sin θ), because part of the gravitational energy must feed rotation. The fraction K^2/R^2 ranks the familiar racers: solid sphere 2/5 → a = (5/7) g sin θ; disc 1/2 → a = (2/3) g sin θ; ring 1 → a = ½ g sin θ. The sphere always wins, the ring always comes last — mass distribution, not mass or size, decides. Note what does *not* appear: M and R cancel, so a big and a small solid sphere tie.

6. The textile and machine thread

Textile machinery is rotating machinery. The **hydroextractor** that spins water out of dyed fabric is a large perforated cylinder whose moment of inertia sets its start-up time and whose imbalance at speed is a standing safety concern; spinning-frame spindles run at tens of thousands of revolutions per minute on tiny I by design; rollers, drums and printing cylinders must be accelerated and braked every cycle, and the torque to do it is τ = Iα with the I of this topic; and the classic flywheel — deliberately built with mass at the rim to *maximise* I per kilogram — steadies the loom and the engine alike. When a machine vibrates, hunts or is slow to start, the engineer's first suspect is usually a moment of inertia someone estimated carelessly.

7. Glossary — key terms of this topic

TERM	MEANING
Moment of inertia (I)	$\Sigma m_i r_i^2$ about a stated axis; a body's resistance to angular acceleration. Unit kg m².
Radius of gyration (K)	The distance at which the whole mass, concentrated there, gives the same I: $I = MK^2$.
Parallel axis theorem	$I = I_{cm} + Md^2$ for any axis parallel to one through the centre of mass.
Perpendicular axis theorem	For a lamina, $I_z = I_x + I_y$ about the point where the axes meet.
Torque (τ)	The rotational analogue of force; $\tau = I\alpha$. Unit N m.
Angular momentum (L)	$L = I\omega$; conserved when no external torque acts.
Rolling without slipping	Rolling in which $v = \omega R$ exactly, with no sliding at the contact point.

8. Key points and summary

- $I = \int r^2\, dm$ depends on mass, its distribution (squared distance) and the chosen axis; $I = MK^2$ defines the radius of gyration.
- The parallel and perpendicular axis theorems generate the standard table: rod $ML^2/12$ (centre) and $ML^2/3$ (end), ring MR^2, disc $\frac{1}{2}MR^2$, solid sphere $(2/5)MR^2$, shell $(2/3)MR^2$.
- Dynamics translates directly: $\tau = I\alpha$, $L = I\omega$, $KE = \frac{1}{2} I\omega^2$; rolling adds rotation's share, giving $a = g \sin\theta / (1 + K^2/R^2)$ down an incline.

9. Practice

Answer in your own words or with full working; brief model answers follow.

1. Two flywheels have the same mass and radius; one is a solid disc, the other a ring. Which has the larger moment of inertia, and by what factor?
2. A thin rod of mass 0.60 kg and length 1.2 m rotates about one end. Find its moment of inertia.
3. Using the perpendicular axis theorem, find the moment of inertia of a disc about a diameter, given $\frac{1}{2} MR^2$ about its central perpendicular axis.
4. A hollow sphere and a solid sphere have equal mass and radius. Their radii of gyration about a diameter are in what ratio?

5. A ring of mass 2.0 kg and radius 0.30 m spins at 10 rad s^{-1} about its central axis. Find its rotational kinetic energy.
6. A solid sphere and a ring are released together down the same incline. State the order in which they arrive at the bottom and each one's acceleration in terms of g sin θ.
7. A torque of 5.0 N m acts on a wheel of I = 2.5 kg m^2. Find the angular acceleration, and the angular speed reached from rest in 4.0 s.
8. Explain, using $I = \int r^2\, dm$, why a hydroextractor's loaded cage (wet fabric pressed to the rim) is harder to start than the same mass of fabric sitting near the hub.

Answer key

1. The ring: MR^2 against the disc's ½ MR^2 — exactly twice as large, because all the ring's mass sits at the full radius.
2. $I = (1/3)\,ML^2 = (1/3) \times 0.60 \times 1.44 = 0.288$ kg m^2.
3. By symmetry the two diameters have equal I, and $I_x + I_y = ½\,MR^2$, so each diameter has I = ¼ MR^2.
4. $K^2 = I/M$: solid sphere $(2/5)R^2$, shell $(2/3)R^2$. Ratio of K values = $\sqrt{(2/5)} : \sqrt{(2/3)} = \sqrt{3} : \sqrt{5} \approx 0.775 : 1$ — the shell's radius of gyration is the larger.
5. $I = MR^2 = 2.0 \times 0.09 = 0.18$ kg m^2; KE = ½ × 0.18 × 100 = 9.0 J.
6. The sphere arrives first. Sphere: $a = g \sin\theta/(1 + 2/5) = (5/7)\, g \sin\theta$; ring: $a = g \sin\theta/(1 + 1) = ½\, g \sin\theta$.
7. $\alpha = \tau/I = 5.0/2.5 = 2.0$ rad s^{-2}; $\omega = \alpha t = 8.0$ rad s^{-1}.
8. Because r enters squared, mass at the rim contributes far more to I than the same mass near the axis; pressed to the rim, the wet load maximises I, so the starting torque ($\tau = I\alpha$) for a given acceleration is greatest — exactly what the integral predicts.
